ECE 253 Course Project Plan: OCR Preprocessing for Phone-Captured Documents

Oct 1, 2026 @Harry

1. Project Overview

This project studies why OCR is inaccurate on casually phone-photographed documents, and uses image-processing methods to bring the accuracy back. Bad shots have many causes; we focus on only the three most common: motion blur, shadow and perspective distortion. We preprocess each defect separately and see how much the recognition error rate drops, and finally study how to order the preprocessing when the three defects are stacked.

Four questions to answer

1. How much does each of the three defects raise OCR's error rates (CER / WER)?
2. How much can traditional image-processing methods recover? How far are they from AI methods?
3. After adapting the existing methods to our self-collected data, are they better than the originals?
4. When the three defects are stacked, what is the most sensible order of preprocessing?

Expected outputs

- A self-collected dataset of phone-photographed documents, with the known original text as ground truth
- Three preprocessing pipelines, each containing at least one traditional method and one comparison method
- OCR comparison results with and without preprocessing, covering both public datasets and the self-collected data
- An order-ablation experiment for preprocessing under the stacking of the three defects
- Proposal, two presentations (11/18–20, 12/5–7), final report (12/11)

2. Handout Requirements and Team Size

First confirm one thing: the handout says groups of 2 or 3, and working alone needs the instructor's or TA's permission first. The team form is due 10/6, so this should be settled this week. If working alone is allowed, the plan below can be used as is; if we are required to form a team, the plan is already cut by artifact, and each person can take one piece.

Handout requirement	How this plan satisfies it
Team of 2–3	Working alone needs approval first (confirm before 10/6)
Each person studies an independent artifact	All three are done, one in depth and the other two to a comparable level (see section 8)
Self-collected data, plus existing datasets	30–50 self-collected document pages under three capture conditions; supplemented by public datasets
At least two methods per artifact, one of them traditional	Two traditional methods per artifact (primary method plus comparison method); AI models only as optional reference lines
Modify existing methods for the self-collected data	At least one adaptive improvement per artifact, compared against the original
Comparison with and without preprocessing	CER / WER as the main metrics, PSNR / SSIM as auxiliary
Discuss the combination of multiple artifacts	Stacking-of-three experiment and preprocessing-order ablation
State each member's contribution percentage	Working alone counts as 100%, stated in the report

3. Technical Plan: Preprocessing the Three Defects

Each defect follows the same routine: first write down the degradation model, then implement one traditional method (the core part written ourselves, not just calling a library), then add a second traditional method as a comparison, and finally make one improvement for the self-collected data. This course is about image processing, so both main lines use traditional techniques, and AI models are only optional reference lines, skipped if time runs short.

3.1 Motion blur

- **Degradation model**: g = h * f + n, where h is the PSF of linear motion (length L, direction θ) and n is noise.
- **Traditional method**: estimate the PSF first, then do Wiener deconvolution. The PSF length and direction are estimated from the dark stripes in the spectrum or from the cepstrum; the Wiener filter is written ourselves in the frequency domain.
- **Comparison method**: also traditional, iterative Richardson-Lucy deconvolution, with the number of iterations tuned as a regularization parameter; if time permits, add a sparse-gradient prior (L0 regularization) for blind deblurring. Wiener is a one-shot frequency-domain solution and RL is the iterative maximum-likelihood solution; the difference between the two in noise amplification and ringing is part of the analysis. AI models such as NAFNet serve only as an optional reference line.

- **Improvement for the self-collected data**: the noise parameter K and the kernel size are not fixed values but set from the text stroke width and the noise level of the self-collected images.
- **Analysis focus**: the effect of the value of K on ringing and noise amplification; how much the result differs when the PSF estimate is slightly off.
- **Course knowledge used**: image degradation models, Fourier transform, inverse filtering, Wiener filtering, PSF estimation, the ringing effect.

3.2 Shadow

- **Degradation model**: I = L × R, i.e. illumination multiplied by reflectance. Shadow belongs to the low-frequency variation of the illumination term, and text belongs to the high-frequency part of the reflectance term.
- **Traditional method**: estimate the background illumination with a large-kernel morphological closing or median filter, divide the original image by the background, then binarize with Otsu or Sauvola adaptive thresholding. The second traditional method is the comparison method below.
- **Comparison method**: homomorphic filtering (high-pass in the log domain) and Retinex (single-scale, multi-scale); like background division they all estimate the illumination term, just in different ways, and they are compared together. Deshadowing networks such as BEDSR-Net serve only as an optional reference line.
- **Improvement for the self-collected data**: the kernel size for background estimation adapts to the text line height rather than being fixed, to avoid a kernel so small that large characters are treated as background and wiped away.
- **Analysis focus**: how the kernel size and the threshold window affect thin strokes; which is harder, the hard shadow cast by a hand or the soft shadow from a lamp.
- **Course knowledge used**: the illumination-reflectance model, homomorphic filtering, morphological operations, Retinex, Otsu / Sauvola thresholds.

3.3 Perspective distortion

- **Degradation model**: when shooting at an angle, the document plane to the image plane is a homography, a 3×3 matrix with 8 degrees of freedom.
- **Traditional method**: Canny edges, then contour polygon approximation to find the four corners of the document, solve the homography matrix, and resample with bilinear or bicubic interpolation to get the frontal view.
- **Comparison method**: another traditional route, fitting the document borders with Hough lines plus RANSAC to solve the homography, or estimating the tilt and perspective from the direction of the text lines (projection profile, Radon transform), which is more robust than the contour-corner method. A homography cannot handle bent paper, so that part is only discussed qualitatively; AI rectification models such as DocTr serve only as an optional reference line.
- **Improvement for the self-collected data**: add a fallback strategy for when corners are blocked by a hand or fall outside the frame, for example extending Hough lines to find intersections, or falling back to

estimating rotation and tilt from the text-line direction.

- **Analysis focus**: the corner-detection failure rate; the effect of the interpolation method on text-edge sharpness; the difference between flat and bent paper.
- **Course knowledge used**: geometric transforms, the homography matrix, edge detection, the Hough transform, image interpolation.

4. Downstream OCR and Combination Experiments

OCR is only an evaluation tool, not the focus of the project. The main recognizer is Tesseract, with its version and parameters (page segmentation mode, language) fixed throughout. PaddleOCR is an optional second recognizer, used to check whether the conclusions hold for more than Tesseract. Tesseract binarizes internally, so the deshadowing result should be tested with two inputs, the grayscale image and the binarized image.

The combination experiment answers the handout's "stacking of multiple artifacts" requirement. When the three defects are stacked, there are 3! = 6 preprocessing orders; run all of them, plus the no-processing baseline, for 7 groups in total.

ID	Preprocessing order	Notes
P0	No processing	Baseline
P1	Deblur → Rectify → Deshadow	Hypothesized optimal: at imaging time illumination comes first, then the camera's geometric projection, and finally blur; restoration goes in reverse order; rectifying first would change the blur kernel, so deblurring goes first
P2	Deblur → Deshadow → Rectify	Exhaustive comparison
P3	Rectify → Deblur → Deshadow	Exhaustive comparison
P4	Rectify → Deshadow → Deblur	Exhaustive comparison, and also the intuitive "straighten the page first" approach
P5	Deshadow → Deblur → Rectify	Exhaustive comparison
P6	Deshadow → Rectify → Deblur	Exhaustive comparison

The hypothesis here was derived from the imaging model and is not necessarily correct; experimental results that disagree with the hypothesis are also valuable findings, and the report should analyze the reasons.

The stacked data has two sources: first, real stacked shots (the combo set); second, synthesis from clean self-collected images in imaging order (add shadow first, then perspective, finally blur), so that the parameters of each stage are known and one can check separately how much error a given step introduced.

5. Data Plan

The self-collected data is a hard requirement of the handout: at least 30 document pages, each shot under four conditions, about 360 images. More pages are fine, with an upper limit of 40 pages, since beyond that one person cannot shoot them all.

Document selection: printed paper pages, book pages, receipts and tables each take a portion, with differences in font size and layout. All are printed from PDFs that have a text layer, so the original text can be extracted directly from the PDF without hand-labeling ground truth, and the images rendered from the PDF can also serve as the clean originals for synthetic data.

Capture condition	Shots per page	How to shoot
Clean baseline	1	Flat, even lighting, as steady as possible, used as this page's reference
Motion blur	3	Hand shake, shooting while walking, fast translation; vary the blur direction and degree as much as possible
Shadow	3	Blocked by a hand, side lighting from a desk lamp, body blocking the light; include both hard and soft shadows
Perspective distortion	3	Tilt angles of about 15°, 30°, 45°, with the page kept flat
Stacking of the three	2	Tilted, shadowed and moving at the same time

Capture conventions: use at least two phones, turn off the document-scanning mode and automatic beautification, and fix the native resolution. Record the phone model, lighting and approximate angle for every image, stored in one metadata CSV. File names are uniformly `docID_condition_seq.jpg`.

Data split: split by page, about one third of the pages for tuning and the rest for the final test; images of the same page must not appear in both, otherwise tuning leaks into the test results.

A limitation to note: the real shots are not pixel-aligned with the clean baseline image, so PSNR / SSIM cannot be computed on real data and only the OCR error rate can be used for evaluation. PSNR / SSIM are used only on synthetic data, where the clean original is known.

Public datasets are used as supplements and comparisons. The candidates are listed below; the download method and license must be verified before deciding.

Type	Candidates	Used for
Phone-captured documents	SmartDoc series	Perspective distortion and overall OCR evaluation
Document geometric distortion	DocUNet-related data	Perspective and bent-paper rectification
Deblurring	Public real-blur datasets, or text-deblurring datasets	Motion blur
Document shadow	Shadow document data from Kligler et al.	Shadow

6. Evaluation Metrics and Experiment Matrix

The main metrics are the OCR character error rate (CER) and word error rate (WER); to judge whether preprocessing helps, look at whether they go down.

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CER = (S + D + I) / N
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Here S, D and I are the numbers of substituted, deleted and inserted characters in the recognition result relative to the original text, and N is the number of characters in the original text. WER uses the same algorithm, with words as the unit.

Auxiliary metrics

- PSNR / SSIM: computed only on synthetic data, to see the quality of the restoration itself
- Corner-detection success rate: used only for perspective rectification, to see whether the corners are found accurately
- Per-page CER distribution: the number of self-collected pages is small, so besides the mean, also plot the per-page distribution to avoid being skewed by individual pages

Experiment matrix: every row must compare with and without preprocessing, and also compare the "original method" with the "version improved for the self-collected data"; this is explicitly required by the handout.

Defect	Comparison groups	Data	Metrics
Motion blur	No processing / Wiener / Wiener (improved parameters) / Richardson-Lucy	Public dataset, self-collected, synthetic	CER, WER; plus PSNR, SSIM on synthetic
Shadow	No processing / background division + Sauvola / homomorphic filtering / Retinex / adaptive-kernel improved version	Public dataset, self-collected, synthetic	CER, WER; plus PSNR, SSIM on synthetic
Perspective distortion	No processing / contour corners + homography / Hough + RANSAC / improved version with fallback strategy	Public dataset, self-collected, synthetic	CER, WER, corner success rate
Stacking of the three	7 orders, P0 to P6	Self-collected combo set, synthetic combo	CER, WER

Every traditional method needs a parameter sweep (for example Wiener's K, the morphological kernel size, Hough's voting threshold), plotting curves of the effect of the parameters on CER; this is the main material for the "discussion" section of the report. If the optional AI reference lines are done, add one more row on the same data and metrics.

7. Staged Plan

The project is cut into seven stages, and each stage ends with a checkable deliverable. The stages are not tied to specific dates: finish one before moving to the next. The course's hard milestones are listed at the end of this section.

Stage 0: Project setup and environment

> **Tasks**: confirm with the instructor or TA that working alone is feasible; complete the sign-up; install Python, OpenCV, scikit-image, Tesseract and a CER / WER computation library; write the evaluation script
>
> **Deliverables**: the reply confirming working alone; an evaluation script that outputs CER and WER for one image
>
> **Completion criteria**: the evaluation script runs through on one test image

Stage 1: Data collection and baseline

Tasks: pick 30 PDF pages and print them; shoot according to the protocol in section 5; organize file names and metadata; download the public datasets; run the OCR baseline without preprocessing

Deliverables: about 360 self-collected images, the metadata CSV, the baseline CER under each condition

Completion criteria: the baseline CER of the blur, shadow and perspective conditions is clearly above the clean baseline, showing that the defects really do hurt OCR; if a condition shows only a small difference, reshoot it

Stage 2: Preliminary experiments and proposal

Tasks: implement a minimal version for each defect (Wiener, background division, contour corners + homography) and confirm on a few images that it runs; write the proposal

Deliverables: three minimal prototypes; the proposal

Completion criteria: all three prototypes lower the CER of at least some images. The handout does not state the proposal's length and format; ask the TA

Stage 3: Primary methods and midterm presentation

Tasks: complete the primary method for each of the three defects; do perspective and shadow first (relatively direct) and leave blur PSF estimation for later; sign up for the midterm presentation per the course schedule; prepare the presentation materials

Deliverables: preliminary results of the primary method for each of the three defects; the midterm presentation slides

Completion criteria: every defect has a "with vs. without preprocessing" CER comparison, even if still rough. This step is also a checkpoint: if it is not met, cut scope in the order of section 8

Stage 4: Midterm presentation

Tasks: present the preliminary results and record the feedback of the instructor and classmates; upload the slides before the deadline as required

Deliverables: the uploaded midterm presentation

Completion criteria: a list of follow-up changes drawn up from the feedback

Stage 5: Comparison methods, improvements and combination experiments

Tasks: implement the second traditional method for each of the three defects; make the improvements for the self-collected data; parameter sweeps; synthetic-data experiments; the P0 to P6 order ablation; at the same time start writing the method and data sections of the report; sign up for the final presentation per the course schedule

Deliverables: complete experiment result tables and curves; a first draft of the method and data chapters of the report

Completion criteria: every row of the experiment matrix in section 6 has numbers

Stage 6: Final presentation and report

Tasks: give the final presentation and upload the slides; then complete the results, discussion, conclusion and contribution statement of the report

Deliverables: final presentation slides; the final report

Completion criteria: the report has all the chapters listed in the handout, and the contribution field states 100%

The course's hard milestones (from the handout; a reminder only, not a schedule)

- 10/6: team form due
- 11/3 11:59 pm: proposal due
- 11/10: midterm presentation sign-up deadline; presentations 11/18–20, upload before 11:59 pm on 11/20
- 11/24: final presentation sign-up deadline; presentations 12/5–7, upload before 11:59 pm on 12/7
- 12/11 11:59 pm: final report due

8. Risks and Scope Cuts

The workload of three defects is large for one person, so the plan keeps a checkpoint before the midterm presentation, and if progress falls behind, scope is cut in the order below. Cutting the scope to two defects still satisfies the handout (each person studies an independent artifact, and the combination is discussed).

Risk	Response
The instructor does not agree to working alone	Confirm before 10/6; if a team is required, split by artifact, each person taking one
The shot defects are too light and the baseline CER shows no difference	In stage 1, first shoot a small batch and run the baseline, then decide whether to make the defects heavier and reshoot
Blur PSF estimation is unstable	First validate the Wiener implementation on synthetic data (known PSF), then move to real images; when the estimate is poor, use the known PSF as an upper bound and say so in the report
Tesseract is inherently poor on some document types (such as receipts)	Report CER separately by document type, not mixed together
A public dataset cannot be downloaded or its license does not allow it	Check before stage 1; if unavailable, substitute self-collected plus synthetic data and explain in the report
Progress falls behind	Check progress before the midterm presentation and cut in the order below

Cut order (cut the earlier ones first)

1. Do not do the AI reference lines
2. Keep only one public dataset per defect
3. Drop the optional advanced methods: L0 blind deblurring, single-scale Retinex
4. Do only the stacking item in the synthetic-data experiments
5. Cut one defect, keeping blur plus shadow or blur plus perspective, whichever has better progress; blur is closest to the course content and is not recommended to be cut

9. Final Report Structure and Contribution Statement

The report chapters follow directly the order listed in the handout, and the material for each chapter is already in this plan. The report is submitted before 11:59 pm on 12/11.

Report chapter	What to write	Source material
Title and members	Project name, authors	Section 1
Introduction and motivation	Why phone-photographed documents are hard to recognize, and how each of the three defects arises	Section 1, the degradation models of section 3
Review of existing methods	Pros and cons of the common methods for each defect	Section 3, supplemented during the literature search in stage 2
Proposed methods	Details of the three preprocessing pipelines, the improvements for the self-collected data, the datasets	Sections 3 and 5
Simulation and comparison	Experimental results, comparison with other methods, discussion	Sections 4 and 6
Conclusion and future work	Conclusions, limitations, directions to continue (such as AI methods, bent paper)	Stage 6
Member contributions	Contribution percentages	Below

Contribution statement: completed alone, the report says 100%. If in the end a team is required, change it to percentages by actual division of labor; the handout says the project grade is weighted by relative contribution.